Control Flows

Why do we need control flow?

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🚦 A traffic light checks conditions: **red** → stop, **green** → go, **yellow** → slow down. Your Python code does the same thing!

🎮 Game

Is health > 0?

→ **Keep playing**

→ **Game over**

🏦 Banking

Is balance enough?

→ **Allow transfer**

→ **Decline**

🌤️ Weather App

Is temp < 10°C?

→ **Show coat icon**

→ **Show sun icon**

What is control flow?

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Control flow structures allow you to dictate the order in which statements execute in your program.

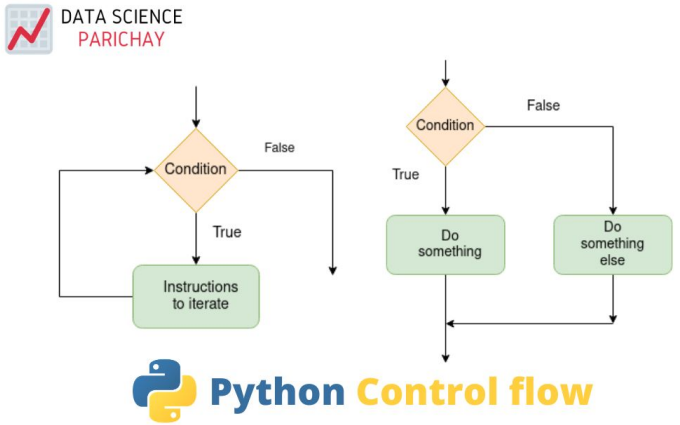
Python manages control flow using 2 core structures

1. **Conditional Statements**

   if / elif / else — branch based on True/False checks.

2. **Loop & Transfer Statements**

   Repeat code and control iteration (covered in a later lesson).



Conditional Statements in control flow

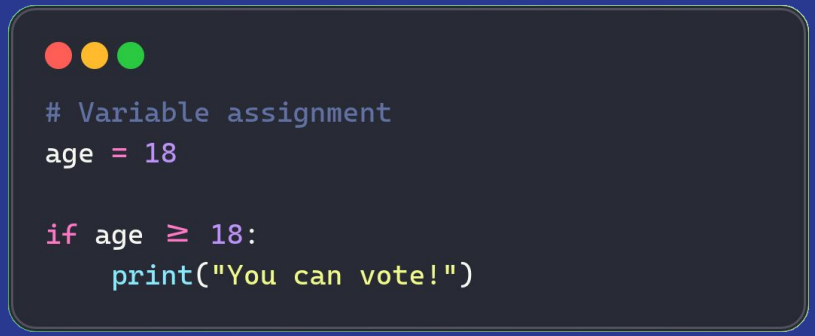
Control Flows: The `if` Statement

Control Flows: The `if` Statement

"If" statements let us control the order code runs in, based on conditions.

- **if** keyword — starts the decision check
- **condition** — an expression that is True or False
- **:** colon — REQUIRED at the end of the line
- **indented block** — only runs if condition is True

Code Sample



Control flow: `else` Statement - A Backup Plan

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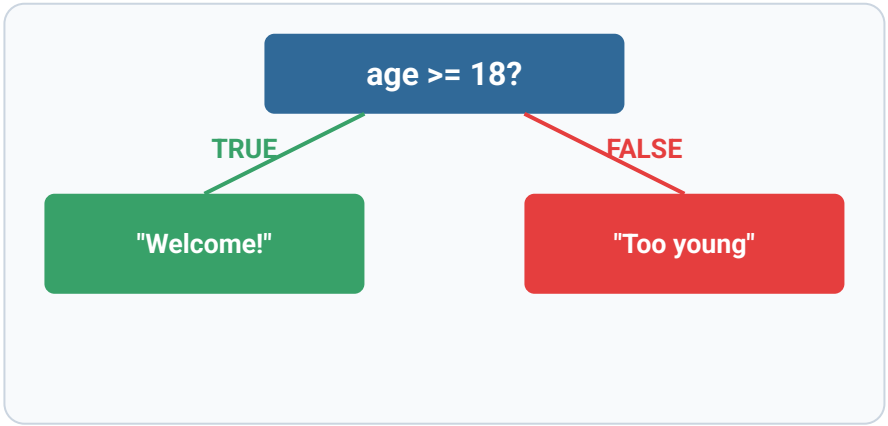
Program to check age:



Code visualizer:

- https://programiz.pro/code-visualizer/python
- https://pythontutor.com/visualize.html#mode=display
- https://learn-code-with-vengleab.vercel.app/

Code Sample

```python
# Current status
age = 15

if age ≥ 18:
    print("Welcome!")
else:
    print("Sorry, too young")
```

Challenge: change age to 20 and re-run — what happens to the output?

Control flow: `elif` Handles Multiple Scenarios

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Grading app:

score >= 90 ? → A — Excellent

score >= 80 ? → B — Good

score >= 70 ? → C — Pass

else (catch-all) → F — Fail

Insight: Python checks conditions top-to-bottom and stops at the FIRST match.

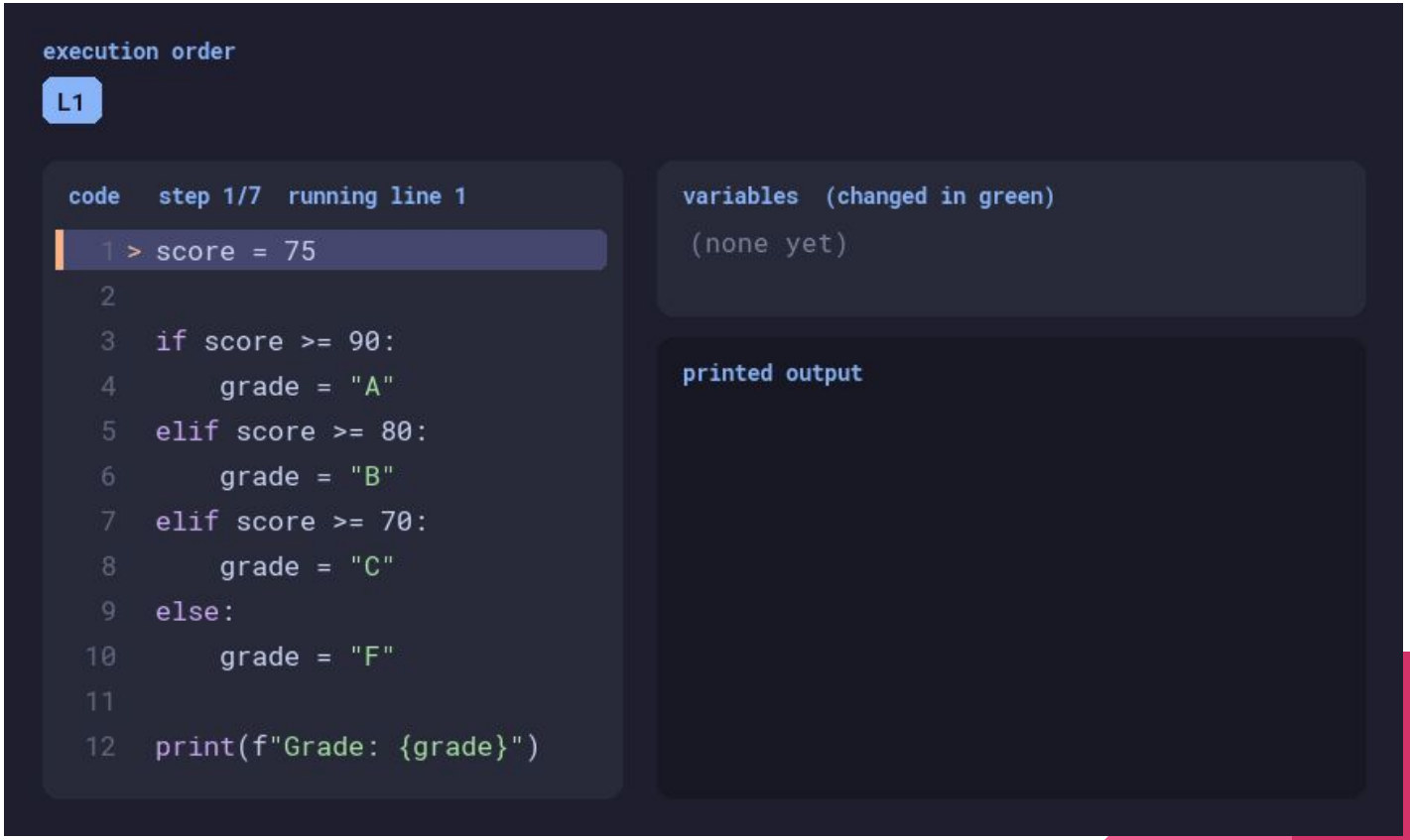
Code Sample

```python
score = 75

if score ≥ 90:
    grade = "A"
elif score ≥ 80:
    grade = "B"
elif score ≥ 70:
    grade = "C"
else:
    grade = "F"

print(f"Grade: {grade}")
```

Visualize code execution



Test your knowledge

🧠 KNOWLEDGE CHECK

What Does This Code Print?

```
x = 7

if x > 10:
    print("Big")
elif x > 5:
    print("Medium")
else:
    print("Small")
```

A) Big

B) Medium

C) Small

D) Nothing — syntax error

🧠 KNOWLEDGE CHECK

What Does This Code Print?

```
x = 7

if x > 10:
    print("Big")
elif x > 5:
    print("Medium")
else:
    print("Small")
```

A) Big

B) Medium

C) Small

D) Nothing — syntax error

Insight: Answer: B) Medium — x is 7, which is > 5 but NOT > 10, so elif matches first.

Nested if Statements

`Nested if` Statements Go Deeper Into Decisions

INDENTATION = NESTING LEVEL

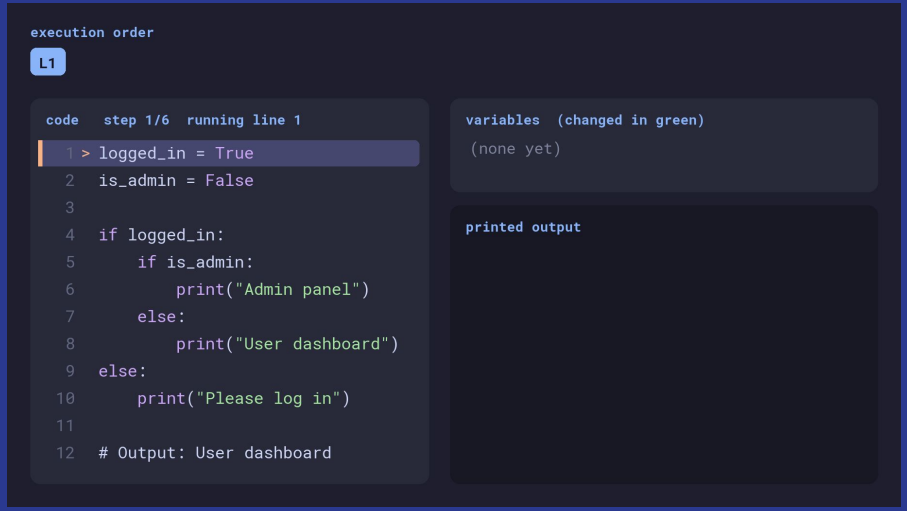
Level 0 — outer if

Level 1 — inner if

Level 2 — inner else

Insight: avoid nesting more than 2–3 levels deep. Consider combining conditions with and / or instead.

Code Example for login check



Ternary Expressions

Ternary Expressions: Simple If-else Into One Line

😵 **Before (4 lines)**

```
if age ≥ 18:
    label = "Adult"
else:
    label = "Minor"
```

→

😎 **After (1 line)**

```
label = "Adult" if age ≥ 18 else "Minor"
```

ANATOMY OF Ternary Expressions

value_if_true if **condition** else **value_if_false**

Insight: use ternary for simple, readable assignments. For complex logic, stick with full if/elif/else.

Practices - Basic If-else

Positive Number & Even Number

TRY IT YOURSELF · LEVEL 1 · EXERCISES 1–2

1 Positive Number

- Ask user to input a number
- If positive, print `Positive`

ex01.py

```
number = float(input("Enter a number: "))

# TODO: check if number is positive
```

Expected output:

[1] number: 10 → Output: Positive
[2] number: -10 → Output:

2 Even Number

- Ask user to input an integer
- If even, print `Even`

ex02.py

```
num = int(input("Enter an integer: "))

# TODO: use % to check even
```

Expected output:

[1] num=10 → Output: even
[2] num= 9 → Output:

Adult Check & Pass or Fail

TRY IT YOURSELF · LEVEL 1 · EXERCISES 3–4

3 Adult Check

- Input age
- 18+ -> `Adult` else `Minor`

ex03.py

```
age = int(input("Enter age: "))

# TODO: compare age to 18
```

Expected output:

[1] age= 18 → Output: Adult
[2] age = 10 → Output: Minor
[3] age= 19 → Output: Adult

4 Pass or Fail

- Input score
- >= 50 -> `Pass`, else `Fail`

ex04.py

```
score = float(input("Enter score: "))

# TODO: compare score to 50
```

Expected output:

[1] score =10 → Output: Fail
[2] score = 50 → Output: Pass
[3] score = 100 → Output: Pass

Bigger Number & Temperature

TRY IT YOURSELF · LEVEL 1 · EXERCISES 5–6

5 Bigger Number

- Ask user to Input two integers
- Print the larger one

ex05.py

```
a = int(input("First number: "))
b = int(input("Second number: "))

# TODO: compare a and b
```

Expected output:

[1] a = 20, b = 20 → Output:
[2] a = 20, b = 10 → Output: 20
[3] a = 18, b = 19 → Output: 19

6 Temperature

- Ask user to Input temperature
- > 30 -> `Hot`, else `Cool`

ex06.py

```
temp = float(input("Enter temperature: "))

# TODO: compare temp to 30
```

Expected output:

[1] temp =10 → Output: Cool
[2] temp = 30 → Output: Cool
[3] temp= 100 → Output: Hot

Divisible by 5 & Username Check

TRY IT YOURSELF · LEVEL 1 · EXERCISES 7–8

7 Divisible by 5

- Ask user to input a number
- Divisible by 5 -> `Yes`, else `No`

ex07.py

```
num = int(input("Enter a number: "))

# TODO: use % 5
```

Expected output:

[1] num = 5 → Output: Yes
[2] num = 10 → Output: Yes
[3] num = 14 → Output: No

8 Username Check

- Ask user to input a username
- `admin` -> `Welcome Admin`, else `Access Denied`

ex08.py

```
username = input("Enter username: ")

# TODO: compare to "admin"
```

Expected output:

[1] username = admin → Output: Welcome Admin
[2] username = Admin → Output: Access Denied
[3] username = 100 → Output: Hot

Practices - Tenerary

Ternary: Maximum & Ternary: Even/Odd

TRY IT YOURSELF · LEVEL 1 · EXERCISES 9–10

9 Ternary: Maximum

ex09.py

```
a = int(input("First: "))
b = int(input("Second: "))

# TODO: one-line ternary
result = ...  # a if ... else b
```

- Use only a ternary operator
- Print the larger number

10 Ternary: Even/Odd

ex10.py

```
num = int(input("Enter integer: "))

# TODO: ternary -> "Even" or "Odd"
```

- Use a ternary operator
- Print `Even` or `Odd`

Practices - Banking and Shopping related

Low Balance Warning & Withdrawal Check

TRY IT YOURSELF · BANKING · EXERCISES 1-2

1 Low Balance Warning

- Ask user to input **balance** and **threshold**
- If balance is **below** threshold, print `Low balance warning`

ex01.py

```
balance = float(input("Enter balance: "))
threshold = float(input("Enter threshold: "))

# TODO: print a warning if balance is below
threshold
```

Expected output:

[1] balance=45, threshold=100 → Output: Low balance warning
[2] balance=150, threshold=100 → Output: (nothing)

2 Withdrawal Check

- Ask user to input **balance** and **withdrawal**
- If withdrawal <= balance print `Approved`, else `Insufficient funds`

ex02.py

```
balance = float(input("Enter balance: "))
withdrawal = float(input("Enter withdrawal: "))

# TODO: print Approved or Insufficient funds
```

Expected output:

[1] balance=200, withdrawal=250 → Output: Insufficient funds
[2] balance=200, withdrawal=150 → Output: Approved

Account Tier & Transfer Verification

TRY IT YOURSELF · BANKING · EXERCISES 3-4

3 Account Tier

- Ask user to input **balance**
- Print the matching tier: Platinum >=10000, Gold >=5000, Silver >=1000, else Basic

ex03.py

```
balance = float(input("Enter balance: "))

# TODO: classify into Platinum / Gold / Silver /
Basic
```

Expected output:

[1] balance=7500 → Output: Gold
[2] balance=800 → Output: Basic

4 Transfer Verification

- First check if account is **verified**
- If verified, check amount against the **daily limit**

ex04.py

```
verified = input("Verified? (yes/no): ") == "yes"
amount = float(input("Enter amount: "))
daily_limit = 5000

# TODO: nested check: verified, then within daily
limit
```

Expected output:

[1] verified=yes, amount=6000 → Output: Flagged for review
[2] verified=no, amount=3000 → Output: Account not verified

Free Shipping & Promo Code

TRY IT YOURSELF · SHOPPING · EXERCISES 1-2

5 Free Shipping

- Ask user to input **cart total**
- If $50 or more, print `You qualify for free shipping!`

ex09.py

```
cart_total = float(input("Enter cart total: "))

# TODO: check if total qualifies for free shipping
```

Expected output:

[1] cart_total=62 → Output: free shipping!
[2] cart_total=30 → Output: (nothing)

6 Promo Code

- Ask user to input **code** and **price**
- Apply 10% discount only if code equals `SAVE10`

ex10.py

```
code = input("Enter promo code: ")
price = float(input("Enter price: "))

# TODO: apply 10% off if code == SAVE10
```

Expected output:

[1] code=SAVE10, price=80 → Output: 72.0
[2] code=X, price=80 → Output: 80

Discount Tiers & Membership Discount

TRY IT YOURSELF · SHOPPING · EXERCISES 3-4

7 Discount Tiers

- Ask user to input **amount spent**
- Print discount tier: 20% >=200, 10% >=100, 5% >=50, else none

ex11.py

```
spent = float(input("Enter amount spent: "))

# TODO: print discount tier
```

Expected output:

[1] spent=120 → Output: 10% off
[2] spent=40 → Output: No discount

8 Membership Discount

- First check if customer **is a member**
- If member, check total against loyalty threshold ($100)

ex12.py

```
is_member = input("Member? (yes/no): ") == "yes"
total = float(input("Enter total: "))

# TODO: nested check: member, then loyalty bonus
```

Expected output:

[1] is_member=yes, total=150 → Output: loyalty discount
[2] is_member=no, total=200 → Output: No discount